

DF13426/Aprile Referee A comments and response

XENON collaboration

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We thank the referee for their time to review the paper and provide constructive comments. To address the first two points, we have divided the results and conclusion section into two sections, facilitating better organization of discussions. The major change in the paper is the modification of section VI (conclusions) to discuss the scope of this paper.

- On the physics motivation of keV signatures out of merger events, refs 14–16 appear to mostly discuss axions at the MeV level rather than keV. Although the black body temperature of neutron stars covers the keV range, it is likely a different picture for merging neutron stars, and in addition, axion and neutrino emission would be quite different from that of X-rays. So my concern of unclear physics motivation for keV level neutrino/axions searches remains. That being said, this is still an interesting work, but the discussions in the conclusion section should be tuned to match the physics motivation.

There is no theoretical motivation to search for keV particles. However, experiments should not be restricted by this lack of theories but explore the unknown regime, especially in search for Beyond Standard Model particles. Now, we have explicitly mentioned the lack of theories in our conclusion.

- On the same point as above, the fact that the neutrino flux values studied in this work exceed the total masses of typical merger systems, as pointed out by the other referee, should be brought up in the conclusion section: this work is an interesting exercise but hasn't probed meaningful parameter spaces yet.

We have made modifications to section VI (conclusions) to explicitly mention and further discuss this point.

- The free electron approximation may be appropriate for MeV neutrino interactions with electrons, but its use for 10s of keV neutrino interactions is questionable, considering that the xenon binding energy is at the same order of magnitude. Even if this is the best assumption the authors can make, the inaccuracy introduced should be discussed.

Our stepping approximation does consider the binding energies of electrons. We agree that there is an inaccuracy of 20% in this approach. To address this point, after equation 2, we have added in the text - "The atomic binding suppresses the cross sections by $\sim 20\%$, as shown in Fig. 2 of [2], using relativistic random phase approximation (RRPA). Since cross section values from RRPA are available only

till 30 keV_{ee}, while our energy region of interest extends to 210 keV_{ee}; we use the stepping approximation — electrons in atomic shells with binding energy lower than the recoil energy are considered to be free.” Since the 20% effect is inconsequential on a log plot, we have not shown an uncertainty band in Fig. 1 (Left) or Fig. 3.

- I found the newly added deadtime discussions informative but also confusing. I imagine that there are not many muons passing through LNGS, so you may have 1 or 0 muons in any 1000s window –if so, how could the muon-induced deadtime distribute uniformly in the window? Similarly the authors claims that 30% of S2-only search data are used as training samples, and the resulted deadtime can only be uniform in the 1000s window if the segments are really small (like seconds)—is this what was used in the analysis? Further, how often does the DAQ system go offline and what is the deadtime due to backgrounds induced?

We have modified the discussion of deadtime to make it more clear. The residual muon flux reaching the experimental hall is $(3.31 \pm 0.03) \cdot 10^{-8} \mu/(\text{cm}^2\text{s})$ [1]. The muon veto water tank has a diameter of 9.6 m and a height of 10.2 m. We observed [323–390] triggers of the muon veto within the 1000 seconds of gravitational wave events, which are uniformly distributed in time, as can be seen in the attached plot. Event duration is defined by the maximum drift time in XENON1T, which is 750 μs , and therefore deadtime due to 30% of S2-only events can be considered uniform. DAQ was unable to record the data [96–228] times in these 1000 second windows, resulting in a deadtime of [2.6–16.7] s. This is referred to as “DAQ busy”, and as can be seen in the attached plot, “DAQ busy” triggers are also uniformly distributed in time. There is a deadtime of $\sim 14\text{s}$ due to background events. The cuts to remove lone S1s and S2s are responsible for rest of the deadtime ($\sim 180\text{s}$ for 2-fold and $\sim 40\text{s}$ for ER, 3-fold and S2-only).

- Table II say the energy region for the 2-fold NR analysis is $\geq 1.6\text{keV}$ but the text says “acceptance ... becoming negligible above 6keV”; this should be reconciled.

Earlier, the energy in Table 2 indicates the threshold cuts on the energy values. We don’t put such an uppercut directly on energy for 2-fold or S2-only analyses, but rather, the cut is placed on the size of S1 and S2 signals. Therefore, the upper values for the energy were absent in Table 2 for these analyses. However, we can make the claims in text based on the attached acceptance plot (acceptance is lower than 1.6% above 6 keV). To avoid any confusion, we have removed the upper value on energies and changed from “energy ranges” to “energy thresholds” in the caption of Table 2 and in the text “The corresponding energy thresholds for these analyses are given in Tab. II”.

- Lastly, a thorough proofreading, especially of the newly added contents, is needed.

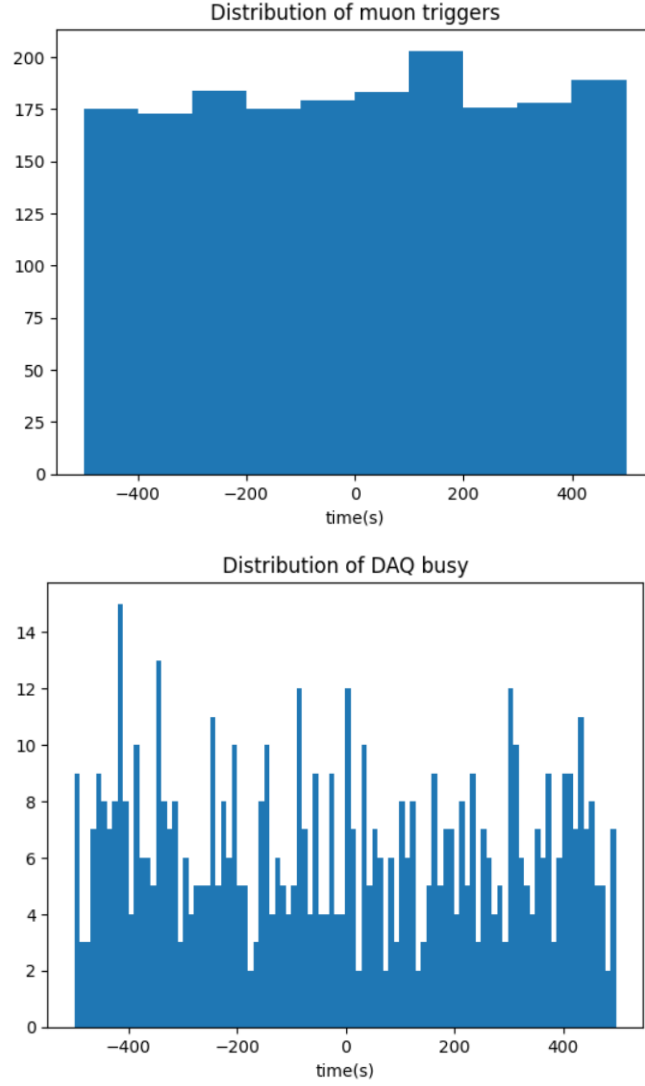


Figure 1: Distribution of the muon veto triggers (top) and DAQ busy triggers (bottom) within 1000 seconds for all the 5 GWs

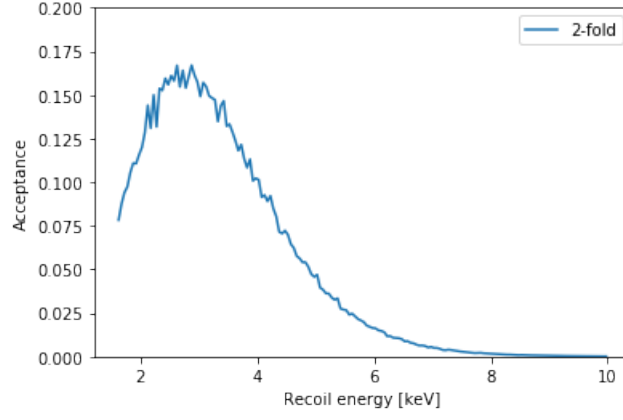


Figure 2: Acceptance for 2-fold analyses

References

- [1] E. Aprile et al. “Conceptual design and simulation of a water Cherenkov muon veto for the XENON1T experiment”. In: *JINST* 9 (2014), P11006. DOI: 10.1088/1748-0221/9/11/P11006. arXiv: 1406.2374 [astro-ph.IM].
- [2] Jiunn-Wei Chen et al. “Low-energy electronic recoil in xenon detectors by solar neutrinos”. In: *Phys. Lett. B* 774 (2017), pp. 656–661. DOI: 10.1016/j.physletb.2017.10.029. arXiv: 1610.04177 [hep-ex].